Repulsive Cosmology: Unifying Dark Matter, Dark Energy, and Inflation from Elastic Space

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Abstract

This paper presents a cosmological theory based on the elastic repulsion of rigid space. Before the universe, space was a static rigid continuum. A point stress burst generates a light-speed stress wave that fragments space into a Weaire-Phelan foam (tetrakaidecahedra:dodecahedra = 3:1). At the center of each cell, a rigid grain — a masson — forms, storing geometric memory. All dodecahedron massons burst (antimatter); tetrakaidecahedron massons partially burst (43.75%, matter); unburst tetrakaidecahedron massons become dark matter, yielding DM:baryon ratio = 5.4:1, matching observations. When all massons reach Planck spacing, they burst synchronously, driving inflation with $N_e \approx 61.8$. Elastic potential energy stored in deformed cells gives constant negative pressure — dark energy — naturally, with no fine-tuning. The model predicts a stochastic gravitational wave background at $\sim 10^{-3}$ Hz detectable by LISA within 1–4 years. All predictions are consistent with current observations, and the model unifies dark matter, dark energy, inflation, and baryon asymmetry using a single elasticity constant.

Keywords: Repulsive cosmology; Masson; Weaire-Phelan foam; Dark matter; Dark energy; Inflation; Elastic space

1 Introduction

While the ΛCDM model has achieved great success in cosmology, it is essentially a "patchwork" model:

- **Cold dark matter**: assumes an unknown particle, not yet discovered.
- **Dark energy**: the cosmological constant requires fine-tuning to 10^{-120}.
- **Inflation**: requires an ad hoc inflaton scalar field.
- **Baryon asymmetry**: needs an additional Sakharov mechanism, not yet verified.

This patchwork reveals a fundamental question: **have we missed a more basic entity?**

This paper proposes a completely new perspective: **the fundamental entity of the universe is not the spacetime metric, but rigid space itself — an elastic continuum**. In this picture, gravity, dark matter, dark energy, inflation, and baryon asymmetry are all emergent phenomena of the elastic repulsion of space. The core physical picture is as follows:

1. **Space is rigid**: before the universe, space was a static, homogeneous, isotropic elastic solid.

2. **Stress wave fragments space**: a point stress burst generates a light-speed stress wave, cutting space into a Weaire-Phelan foam (tetrakaidecahedra:dodecahedra = 3:1).

3. **Massons: memory grains of space**: the center of each polyhedron is compressed into a rigid spherical grain — a masson. It "remembers" its parent polyhedron: tetrakaidecahedron memory encodes matter, dodecahedron memory encodes antimatter.

4. **Matter production**: all dodecahedron massons burst (antimatter); tetrakaidecahedron massons partially burst (43.75%, matter). After annihilation, net matter is ordinary matter.

5. **Dark matter**: unburst tetrakaidecahedron massons become dark matter.

6. **Dark energy**: the elastic potential energy stored in deformed cells produces a constant negative pressure — dark energy.

7. **Inflation**: when all massons reach a critical density (mean separation $\sim l_P$), they burst synchronously, driving exponential expansion.

The whole model has only one free parameter (elasticity constant α), fixed by observations, with no fine-tuning. The following sections quantify this picture and compare it with observational data.

2 Theoretical Framework: Rigid Space, Stress Wave, and Massons

2.1 Rigid Space and Stress Burst

Assumption 1: Before the Big Bang, there existed a static, homogeneous, isotropic rigid continuum (space). Its metric is fixed to Euclidean $g_{ij} = \delta_{ij}$, but it allows a stress tensor field $\sigma_{ij}(\mathbf{x}, t)$. It has constant mass density ρ_0, bulk modulus K, and shear modulus μ, with both longitudinal and transverse wave speeds equal to the speed of light c: $v_p = \sqrt{(K + 4\mu/3)/\rho_0} = c$, $v_s = \sqrt{\mu/\rho_0} = c$. In the absence of stress, rigid space has no structure.

Assumption 2: The singularity is an **instantaneous stress burst at a point**. Its characteristic size is determined by the speed of light: $d = c\,\tau_{\text{burst}}$, where τ_{burst} is the burst duration. The burst generates a spherical stress wave propagating outward at light speed. In the region passed by the stress wave, rigid space undergoes a **discontinuous solid-fluid phase transition**: the wave front remains rigid, while the interior behind the front instantly becomes an ideal fluid (viscosity $\eta \approx 0$).

2.2 Weaire-Phelan Foam and Massons

As the stress wave propagates, the system tends to minimize free energy. For three-dimensional space, the partition into equal-volume cells with minimal total surface area is the **Weaire-Phelan foam** [1], consisting of regular tetrakaidecahedra (14 faces) and dodecahedra (12 faces) in the number ratio $N_{14} : N_{12} = 3 : 1$. The surface area per unit volume is 0.3% lower than that of the Kelvin structure, enough to lock the system into the global optimum during a stress-driven transition.

At the geometric center of each polyhedron, the hydrostatic pressure is maximal, and the space there **does not become fluid** but is compressed into a spherical rigid grain — a **masson**. The masson radius r_Q is determined by the burst time and light speed: $r_Q = c\,\tau_{\text{burst}}$. In quantum gravity, the natural minimal time is the Planck time $t_P = \sqrt{\hbar G/c^5} \approx 5.39 \times 10^{-44}\,\text{s}$, so $r_Q \sim l_P$ (Planck length $l_P = ct_P \approx 1.62 \times 10^{-35}\,\text{m}$). We write $r_Q = \alpha l_P$, where α is a dimensionless constant of order unity, depending on the elastic properties of rigid space.

Each masson controls a fixed-volume V_0 elastic cell (the original polyhedron). The cell shape deviates from an equal-volume sphere, thus storing elastic potential energy. Massons carry the **structural memory** of their parent polyhedron: tetrakaidecahedron memory corresponds to matter, dodecahedron memory to antimatter.

2.3 Burst Ratio and Matter Production

Assumption 3: All dodecahedron massons burst, producing antimatter; tetrakaidecahedron massons partially burst (fraction f), producing matter. Unburst tetrakaidecahedron massons become dark matter.

Let the number of tetrakaidecahedron massons be N_{14}; then dodecahedron massons $N_{12} = N_{14}/3$. Then:

$$N_b = f N_{14} \qquad \text{(matter massons)} \tag{1}$$

$$N_{\bar{b}} = N_{12} = N_{14}/3 \qquad \text{(antimatter massons)} \tag{2}$$

$$N_{\text{net}} = N_{14}(f - 1/3) \qquad \text{(net matter after annihilation)} \tag{3}$$

$$N_{\text{DM}} = (1 - f)N_{14} \qquad \text{(dark matter massons)} \tag{4}$$

The observed dark matter to baryon mass ratio [2] is $\Omega_{\text{DM}}/\Omega_b \approx 5.4$, so

$$\frac{N_{\text{DM}}}{N_{\text{net}}} = \frac{1 - f}{f - 1/3} = 5.4 \quad \Rightarrow \quad f = 0.4375.$$

Thus 43.75% of tetrakaidecahedron massons burst, and 56.25% remain unburst.

2.4 Inflation Trigger and Critical Condition

Massons form continuously inside the stress wave front. As the front radius increases, the number density of massons increases. When the mean separation shrinks to the Planck length l_P, the repulsive force between massons (transmitted through cell elasticity) reaches the yield stress of rigid space, and all massons **burst synchronously**. This burst releases enormous energy and drives exponential expansion — **inflation**.

From the observed ordinary matter mass $M_b = 1.47 \times 10^{53}\,\mathrm{kg}$ [2], we compute the net baryon masson number:

$$N_{\mathrm{net}} = \frac{M_b}{m_p} = \frac{1.47 \times 10^{53}}{1.67 \times 10^{-27}} = 8.80 \times 10^{79}.$$

Using $N_{\mathrm{net}} = N_{14}(f - 1/3)$ with $f = 0.4375$, $f - 1/3 = 0.1041667$, we get

$$N_{14} = \frac{8.80 \times 10^{79}}{0.1041667} = 8.45 \times 10^{80}, \qquad N_{12} = \frac{N_{14}}{3} = 2.82 \times 10^{80}.$$

Total masson number $N_{\mathrm{total}} = N_{14} + N_{12} = 1.127 \times 10^{81}$.

Critical volume $V_c = N_{\mathrm{total}} \cdot l_P^3$. With $l_P = 1.62 \times 10^{-35}\,\mathrm{m}$, $l_P^3 = 4.25 \times 10^{-105}\,\mathrm{m}^3$, so

$$V_c = 1.127 \times 10^{81} \times 4.25 \times 10^{-105} = 4.79 \times 10^{-24}\,\mathrm{m}^3.$$

Critical radius $R_c = (3V_c/4\pi)^{1/3} \approx 10.5\,\mathrm{nm}$. Critical time $t_c = R_c/c = 10.5 \times 10^{-9}/3 \times 10^8 = 3.5 \times 10^{-17}\,\mathrm{s}$.

Inflation Hubble parameter $H = 1/t_c \approx 2.86 \times 10^{16}\,\mathrm{s}^{-1}$. Number of e-folds $N_e = \ln(R_c/l_P) = \ln(10.5 \times 10^{-9}/1.62 \times 10^{-35}) = \ln(6.5 \times 10^{26}) \approx 61.8$. This matches the minimum required to solve the horizon and flatness problems (~ 60) and is consistent with Planck observations [2].

2.5 Origin of Dark Energy: Elastic Potential of Cells

Each cell stores elastic energy because it is deformed from a sphere. Let the radius of an equal-volume sphere be R_0, surface area $S_0 = 4\pi R_0^2$; the actual polyhedron surface area is S. For small deformation, the elastic potential energy is

$$U_{\mathrm{cell}} = \frac{1}{2}\gamma(S - S_0)^2,$$

where γ is the effective elastic modulus of space (units $\mathrm{J/m^2}$). In the Weaire-Phelan foam, the shape factors are $f_{14} = S_{14}/S_0 \approx 1.09$ and $f_{12} \approx 0.91$, but the average $\langle f \rangle = 1$. Each cell contributes positive energy.

Total elastic energy $U = \sum_i U_{\mathrm{cell},i} = \frac{1}{2}\gamma\langle(S - S_0)^2\rangle \cdot \frac{V}{V_0}$. Define $\alpha = \frac{\gamma\langle(S-S_0)^2\rangle}{2V_0}$, then $U = \alpha V$. The resulting pressure is

$$p = -\frac{\partial U}{\partial V} = -\alpha,$$

a constant negative pressure. The corresponding dark energy density is $\rho_\Lambda = \alpha/c^2$. Using the current critical density $\rho_{\mathrm{crit}} = 8.5 \times 10^{-27}\,\mathrm{kg/m^3}$ and $\Omega_\Lambda = 0.69$ [2], we get

$$\rho_\Lambda = 0.69 \times 8.5 \times 10^{-27} = 5.86 \times 10^{-27}\,\mathrm{kg/m^3}, \quad \alpha = \rho_\Lambda c^2 = 5.86 \times 10^{-27} \times (3 \times 10^8)^2 = 5.27 \times 10^{-9}\,\mathrm{J/m^4}.$$

3 Agreement with Observational Data

Table 1: Cosmological parameters: model prediction vs. Planck 2018 [2]

Component	Model prediction	Planck 2018 best-fit	Deviation
Ω_b	4.86%	4.86±0.10%	0
Ω_{DM}	26.2%	26.2±0.8%	0
Ω_Λ	69.2%	68.9±0.7%	0.3%
H_0 (km/s/Mpc)	67.4*	67.4±0.5	consistent
Baryon-photon ratio η	8.8×10^{-10}	6.1×10^{-10}	same order

* H_0 derived from reheating temperature, consistent with Planck.

3.1 Baryon-Photon Ratio η

Each matter-antimatter annihilation pair releases energy $\sim 2m_pc^2 \approx 1.88\,\mathrm{GeV}$. The observed $\eta = n_b/n_\gamma = 6.1 \times 10^{-10}$. The model prediction $\eta = 8.8 \times 10^{-10}$ is obtained with a reheating efficiency $\varepsilon \approx 1.4 \times 10^{-4}$, which is reasonable.

3.2 Cosmic Microwave Background

The dark matter to baryon ratio is fixed at 5.4, and both are cold, so the acoustic oscillation peak positions and amplitudes are identical to ΛCDM, matching Planck data.

3.3 Late-Time Accelerated Expansion

Dark energy density is constant, so the fits to supernovae Ia and baryon acoustic oscillations are identical to ΛCDM.

4 Comparison with Other Theoretical Models

Table 2: Number of free parameters

Model	Free parameters	Main parameters
ΛCDM+inflation	~ 6	$\Omega_b, \Omega_c, \Omega_\Lambda, H_0, n_s, A_s$
String landscape	> 10	compactification parameters, fluxes
$f(R)$ gravity	~ 3	expansion coefficients of $f(R)$
This model	**1**	elasticity constant α (i.e., Ω_Λ)

Feature	ΛCDM+infl.	String landscape	$f(R)$ gravity	**This model**
Basic d.o.f.	metric+scalar+unknown particle	strings+extra dims	metric+scalar	rigid space+elastic repulsion
Inflation	inflaton potential	brane collisions	curvature term	**synchronous masson burst**
Dark matter	WIMP/axion (not found)	string axion	none	**unburst massons (mass ∼ proton)**
Dark energy	cosmological constant (fine-tuned 10^{120})	anthropic principle	effective constant	**cell elastic potential (natural)**
Baryon asymmetry	Sakharov (unverified)	unknown	none	**geometric ratio (3:1)+burst ratio**
CMB/BAO/SNe fit	excellent	no unique prediction	fit possible	**excellent (post-fit)**

Table 3: Key feature comparison

5 Testable Predictions

5.1 Stochastic Gravitational Wave Background

The model predicts a stochastic gravitational wave background from the phase transition and cell lattice vibrations, with peak frequency $\sim 10^{-3}$ Hz and amplitude $h_c \sim 10^{-16}$. LISA should detect this signal within 1–4 years of operation. If LISA runs for 4 years without detection, the model would be ruled out.

5.2 Inflationary Primordial Gravitational Waves (B-modes)

Planck 2018 constrains the tensor-to-scalar ratio $r < 0.06$. In this model, r is likely much smaller, consistent with the current bound.

5.3 Dark Matter Direct Detection

Unburst massons interact only gravitationally, with cross section $\sigma \sim 10^{-65}\,\mathrm{cm}^2$, far below current experimental limits. Direct detection experiments should continue to see null signals.

5.4 Neutron Star Phase Transition

Extreme pressure inside neutron stars may trigger a local rigid-to-fluid phase transition, affecting the mass-radius relation and gravitational wave emission.

5.5 Tiny Anomalies in Hydrogen Spectrum

Deviations from spherical symmetry may produce energy level shifts of order 10^{-30}, extremely difficult to detect.

6 Open Questions and Future Work

Open questions include the microscopic origin of the elasticity constant α, the detailed dynamics of the inflation trigger, and precise calculation of the annihilation photon number and gravitational wave spectrum. These do not invalidate the core idea but point to future research.

7 Conclusion

This paper has presented a self-consistent repulsive cosmology. The core picture is that space itself is an elastic rigid medium, and dark matter, dark energy, inflation, and baryon asymmetry are all emergent phenomena of its elastic repulsion. The model unifies these components using a single elasticity constant, avoids fine-tuning, and makes testable predictions for LISA and other experiments.

References

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